

Kenney Tran

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Portfolio: kenney-personal.vercel.app

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Education

- **Boston University** Boston, MA
• *Bachelor of Science in Computer Science (GPA: 3.2)* *Expected Graduation: 2026*
 - Relevant Coursework: Data Structures & Algorithms, Database Systems, Fullstack Development, Machine Learning, Data Science, ML for Business Analytics
 - Transferred from UMass Boston (Computer Engineering) in 2023

Technical Skills

- **Programming Languages:** HTML, CSS, JavaScript/TypeScript, Bash, Python, Java, C, SQL
- **Frameworks & Libraries:** React, Next.js, Node.js, Express, Spring Boot, Flask, MUI, Tailwind CSS, Firebase, scikit-learn, NumPy, pandas, Matplotlib
- **Databases & Caching:** MongoDB, Redis, Firebase Firestore
- **Developer Tools:** Git, GitHub, Vercel, Postman, Google Cloud Platform

Projects

- **BU Transit Tracker Web Application (Next.js, TypeScript, Node.js, Redis, Google Maps API, Vercel):**
 - Built **full-stack real-time bus tracking application** integrating BU TransLoc API and Google Maps to display live positions, routes, and ETAs for **3+ bus lines**
 - Implemented **Redis caching layer** reducing external API calls by **60%** and latency to **under 100ms** for high-frequency updates
 - Deployed on **Vercel with CI/CD pipeline** for automated builds/deployments, achieving **99.9% uptime**
- **Discord Music Bot (TypeScript, Node.js, Discord.js, yt-dlp, FFmpeg, Opus):**
 - Engineered **high-performance Discord music bot** with real-time YouTube audio streaming using **yt-dlp and FFmpeg Opus encoding**, eliminating disk I/O overhead
 - Implemented **piped process streams** with optimized buffering achieving **sub-200ms latency** for playback across multiple concurrent servers
 - Designed **FIFO queue system** with slash command interface (/play, /skip, /queue) and auto-reconnection logic ensuring **99%+ uptime**
- **Airbnb Price Prediction Model (Python, scikit-learn, pandas, NumPy, Matplotlib, XGBoost):**
 - Designed **supervised ML pipeline** predicting Airbnb prices across **76K+ records**, achieving **$R^2 = 0.71$** with Random Forest and reducing RMSE by **35%**
 - Performed **feature engineering on 20+ attributes** using imputation and log transformations to handle **15% missing data** and skewed distributions
 - Compared **5 baseline models** with tree-based ensembles and tuned hyperparameters using GridSearchCV for optimization
- **Multi-threaded Image Processing Server (C, POSIX Threads, Sockets, Semaphores):**
 - Built **thread-safe C server** with FIFO queue and **8 worker threads** handling image operations with **5x throughput improvement**
 - Used **mutexes and semaphores** to eliminate race conditions, improving stability under **100+ concurrent requests**
 - Implemented socket-based communication supporting **3 image operations** with request parsing, response formatting, and graceful shutdown